

The Future is Here: Verbal Interaction with NPCs on Mobile

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Figure 1: Verbal interactions with the NPC. Subfigure (a) show it holding a consistent conversation, (b) protesting when asked to fetch an unreachable object, (c) using a Multilayer Perceptron (MLP) network with Unity ML-Agents to push the chair.

ABSTRACT

Applying large Language Models (LM) to unlock natural language based interactions with game NPCs is an emerging frontier. But the LMs of today are prohibitively expensive to run locally, especially on mobile platforms due to limited runtime resources, thermal issues and strong FPS requirements. In this work, we identified a series of mobile-friendly small LMs called TinyStories[Eldan and Li 2023] that can generate simple but coherent English. We repurposed this model to generate NPC dialogues. We then combined it with Unity ML-Agents, creating an NPC that can perform complex actions and interact verbally with the player. Our implementation runs on current-generation mobile platforms, at the level of performance required for real-time gaming. With this work, we offer a glimpse into the future of gaming: interacting with complex NPCs using natural speech. We also demonstrate it is possible to run LMs alongside Unity ML-Agents on mobile with great results. This will significantly enhance the accessibility and interactivity in games. Lastly, we want to attract academia and industry attention to the area of mobile / edge friendly LMs, as we believe this will be a fertile ground for exploration.

CCS CONCEPTS

• Computing methodologies → Speech recognition; • Human-centered computing → Smartphones.

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KEYWORDS

user interaction in games, speech-based user interaction, speech recognition, language model, unity ml-agents, mobile games

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1 INTRODUCTION

Unity ML-Agents toolkit is an open source project that enables training and deploying ML-based agents in diverse Unity environments [Juliani et al. 2020]. The toolkit integrates with Unity's own deep learning inference engine, Sentis which provides on-device acceleration. Arm®'s past contributions to Unity ML-Agents project, with a focus on mobile[Mitsunami 2023], serve as a basis for this work.

The predominant deployment method for large Language Models (LM) is cloud-based, due to the sheer size of such models. This is unsuitable for interactive gaming, where low latency is crucial. We identified a series of compact LMs - TinyStories, which were part of an investigation into the smallest possible LM that can still generate coherent English. A novel knowledge distillation method was employed, to train models on diverse short stories, generated by larger LMs, with the vocabulary of a typical 3-4 year old child. As a result, this work has produced a series of small LMs, with the number of parameters ranging from 1M to 33M.

2 DESIGN AND IMPLEMENTATION

Our NPC can: (1) Follow player's verbal commands in natural language, to perform complex actions; and (2) hold an interactive conversation. The pipeline is composed of 5 modules (see figure 2).

The language module: TinyStories generates the NPCs interactive dialogue. Originally, this model was trained only with the purpose of generating short stories instead of conversations. We found that by framing the player’s interactions as part of a story, we can prompt the model into producing the desired replies.

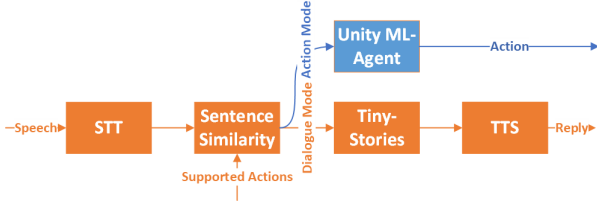


Figure 2: System diagram showing the full pipeline. STT: Speech To Text. TTS: Text To Speech.

We call this technique storification. The process involves: (1) Providing an initial context that reads like the beginning of a story. (2) Embedding player input into the initial context as the player’s dialogue. (3) Generating an output from the LM, from which the NPC’s reply can be extracted. The TinyStories-33M variant with greedy search generation strategy was chosen. We also implemented a memory cache so that the NPC can remember the context. In addition, we utilized the Sentis’ optimized CPU backend, and cached past key-value embeddings to boost runtime performance.

The decision module uses Sentence Similarity, which discerns if the player’s input speech is a command or an invitation to chat. We opted for the Sentence similarity matching method [Reimers and Gurevych 2019]. We represent the list of actions as a list of key sentences, and then compare the player’s input to this list. On a matched sentence, the NPC enters the action mode. Otherwise, it defaults to the dialogue mode. We used the sentence transformer all-MiniLM-L6-v2 model to compute sentence embeddings. A cosine similarity between a pair of embeddings is calculated and compared against a threshold to determine a match.

The action module uses a Multilayer Perceptron (MLP) with Unity ML-Agents which carries out the selected actions. An example action is to, in a physical environment, push a chair to a named location. We chose, for this case, an MLP with 2 hidden layers of 256 units and trained using the Proximal Policy Optimization (PPO) algorithm for a total of 4.2 million steps. Our observations include an array of ray-cast sensors attached to the NPC, and knowledge of the chair and target positions.

Speech to Text (STT) and Text to Speech (TTS) modules convert the player speech to text, and synthesize the LM’s responses to speech. We used the Android SpeechRecognizer and Android TextToSpeech APIs respectively.

3 EVALUATION AND CONCLUSION

The NPC’s capabilities are shown in Figure 1. The real-time performance of the NPC was evaluated using these two metrics:

- (1) Average initial reaction time: see Table 1. This refers to the duration from when the NPC first receives an input until the start of its response, either through an action or a reply. A long initial reaction time implies a long delay before seeing any response from the NPC.
- (2) Reply time per token: 14.8 ms (dialogue mode only), which is the time taken to generate each subsequent token. A long reply time per token indicates that the reply is choppy and laggy.

These evaluations were performed on a Google Pixel™ 8 Pro with Nona-core (1x3.0 GHz Arm® Cortex®-X3 and 4x2.45 GHz Cortex®-A715 and 4x2.15 GHz Cortex®-A510).

Table 1: Average Initial Reaction Time

Stage	Action(ms)	Dialogue (ms)
Sentence Similarity	30.0	25.2
TinyStories 33M	-	59.7
Misc.	2.5	35
Overall Reaction	32.5	119.9

In conclusion, we repurposed a small LM: TinyStories, and combined it with Unity ML-Agents to bring an NPC that can perform complex actions and hold an interactive conversation. The implications of our work are:

- (1) Boost accessibility and interactivity in gaming by ways of verbal interaction which humanizes and enlivens the NPC.
- (2) Empower developers to create richer gaming experiences by exploring new ways to utilize Unity ML-Agents’ easy tooling.
- (3) Highlight the growing significance of LMs in gaming and technology. For the first time, LMs afford us an intuitive and natural way of interfacing with computers.
- (4) Inspire innovation in compact LMs. To democratize LMs, we need more research and industry efforts on mobile friendly LMs.

Notably, we demonstrated successful deployment of this combined system locally on mobile hardware. We hope that our work not only highlights the usefulness of having more mobile friendly LMs, but also puts a spotlight on an area with untapped potential for further research.

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